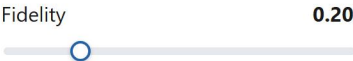


MISSION PROFILE

Performance

Balanced

Safety



CONSTRAINTS

- ☐ Regulatory certification
- ☒ Real-time inference
- ☒ Sensor drift expected

RECOMMENDED

OURS

Unified Trust Score

0.80

40.8 W

0.64

+14.2%

MAE

τ

$\Delta@20\%$

Most accurate; Pareto-dominates every baseline.

Minimal degradation under sensor noise.

Consensus flags payload as the dominant power driver.

Deploy model

MODEL RANKING

profile: performance-critical

1	OURS	P		0.80
2	CatBoost			0.76
3	LightGBM			0.69
4	GradBoost			0.69
5	RandForest			0.62
6	Ridge	P		0.40
7	XGBoost			0.62

Trust axes are min-max normalised across the model pool (higher is better on each); the Unified Trust Score is their weighted sum. Constraints act as hard filters. Numbers reproduce the paper's held-out flight-telemetry results.